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SYSTEMS, APPARATUSES, AND METHODS FOR LEAK DETECTION USING A CAPILLARY TUBE

Abstract

In some examples, a leak detection apparatus is described herein. In one or more example embodiments, the leak detection apparatus includes a spill collector configured to collect a liquid of interest from a leakage of the liquid of interest. In one or more examples, the leak detection apparatus alternatively or additionally includes a capillary tube a first end configured to immerse in the liquid of interest in the spill collector; and a second end configured to mechanically couple with a pressure sensor, the pressure sensor configured to measure an air pressure in the capillary tube. In one or more examples, the leak detection apparatus alternatively or additionally includes one or more computing devices electronically coupled with the pressure sensor, the one or more computing devices configured to distinguish the liquid of interest from another liquid using the measured air pressure in the capillary tube.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority pursuant to 35 U.S.C. 119(a) to Indian Application No. 202411015947, filed Mar. 6, 2024, which application is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Liquid leak detection apparatuses may be used to detect various types of liquid leakages, for example leakage of a coolant of a battery pack. Some detection apparatuses use resistive sensors to detect a leakage, such as a coolant leakage. However, such detection apparatuses may suffer from false alarms. For example, they may detect water condensation as a coolant leakage. Through applied effort, ingenuity, and innovation, many of these identified problems have been solved by developing solutions that are included in embodiments of the present disclosure, examples of which are described in detail herein.

BRIEF SUMMARY

[0003] In some examples, a leak detection apparatus is described herein. In one or more example embodiments, the leak detection apparatus includes a spill collector configured to collect a liquid of interest from a leakage of the liquid of interest. In one or more examples, the leak detection apparatus alternatively or additionally includes a capillary tube a first end configured to immerse in the liquid of interest in the spill collector; and a second end configured to mechanically couple with a pressure sensor, the pressure sensor configured to measure an air pressure in the capillary tube. In one or more examples, the leak detection apparatus alternatively or additionally includes one or more computing devices electronically coupled with the pressure sensor, the one or more computing devices configured to distinguish the liquid of interest from another liquid using the measured air pressure in the capillary tube.

[0004] In one or more examples, the leak detection apparatus alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to determine a signature parameter for the liquid of interest, wherein the signature parameter comprises a viscosity difference between the liquid of interest and the other liquid and is determined using the measured air pressure in the capillary tube; and distinguish the liquid of interest from the other liquid using a capillary rise differentiation of the liquid of interest from the other liquid, wherein the capillary rise differentiation of the liquid of interest from the other liquid is based on the signature parameter.

[0005] In one or more examples, the leak detection apparatus alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to compare the measured air pressure in the capillary tube with an air pressure threshold; and determine a presence of the liquid of interest when the measured air pressure in the capillary tube is above the air pressure threshold.

[0006] In one or more examples, the leak detection apparatus alternatively or additionally includes a temperature sensor, wherein the temperature sensor comprises a temperature probe configured to engage with at least one of the liquid of interest or the other liquid in the spill collector, the temperature sensor configured to measure a temperature in the spill collector using the temperature probe.

[0007] In one or more examples, the leak detection apparatus alternatively or additionally includes

one or more computing devices, the one or more computing devices are further configured to vary the air pressure threshold using the measured temperature.

[0008] In one or more examples, the leak detection apparatus alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to increase the air pressure threshold when the measured temperature is above a temperature threshold; and decrease the air pressure threshold when the measured temperature is below the temperature threshold.

[0009] In one or more examples, the leak detection apparatus alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to generate an alarm when the presence of the liquid of interest is determined.

[0010] In one or more examples, the liquid of interest is a coolant of a battery pack, the spill collector is a bottom cover of the battery pack, and wherein the other liquid is at least one of water generated by condensation, melted wax, or melted adhesive in the spill collector.

[0011] In some examples, a coolant leak detection system is described. In one or more example embodiments, the coolant leak detection system includes a spill collector configured to collect a coolant leaked from a battery pack. In one or more example embodiments, the coolant leak detection system alternatively or additionally includes a capillary tube including a first end configured to immerse in the coolant when leaked in the spill collector; and a second end configured to mechanically couple with a pressure sensor, the pressure sensor configured to measure an air pressure in the capillary tube above the coolant. In one or more example embodiments, the coolant leak detection system alternatively or additionally includes one or more computing devices electronically coupled with the pressure sensor, the one or more computing devices configured to distinguish the coolant from another liquid in the spill collector using the measured air pressure in the capillary tube.

[0012] In one or more example embodiments, the coolant leak detection system alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to determine a signature parameter for the coolant, wherein the signature parameter comprises a viscosity difference between the coolant and the other liquid and is determined using the measured air pressure in the capillary tube; and distinguish the coolant from the other liquid using a capillary rise differentiation of the coolant from the other liquid, wherein the capillary rise differentiation of the coolant from the other liquid is based on the signature parameter.

[0013] In one or more example embodiments, the coolant leak detection system alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to compare the measured air pressure in the capillary tube with an air pressure threshold; and determine a presence of the coolant when the measured air pressure in the capillary tube is above the air pressure threshold.

[0014] In one or more example embodiments, the coolant leak detection system alternatively or additionally includes a temperature sensor, wherein the temperature sensor comprises a temperature probe configured to engage with at least one of the coolant or the other liquid in the spill collector, the temperature sensor configured to measure a temperature in the spill collector using the temperature probe.

[0015] In one or more example embodiments, the coolant leak detection system alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to vary the air pressure threshold using the measured temperature.

[0016] In one or more example embodiments, the coolant leak detection system alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to increase the air pressure threshold when the measured temperature is above a temperature threshold; and decrease the air pressure threshold when the measured temperature is below the temperature threshold.

[0017] In one or more example embodiments, the coolant leak detection system alternatively or additionally includes one or more computing devices, the one or more computing devices are further configured to generate an alarm when the presence of the coolant is determined.

[0018] In some examples, the coolant is a coolant of a battery pack and the spill collector is a bottom cover of the battery pack, and wherein the other liquid is at least one of water generated by condensation, melted wax, or melted adhesive in the spill collector.

[0019] In one or more example embodiments, a leak detection method is described herein. In some example embodiments, the leak detection method includes measuring, by a pressure sensor, an air pressure in a capillary tube, wherein a first end of the capillary tube is configured to immerse in a liquid of interest in a spill collector configured to collect the liquid of interest from a leakage of the liquid of interest, and wherein the pressure sensor is mechanically coupled to a second end of the capillary tube. In some example embodiments, the leak detection method alternatively or additionally includes distinguishing, by one or more computing devices electronically coupled with the pressure sensor, the liquid of interest from another liquid using the measured air pressure in the capillary tube.

[0020] In some example embodiments, the leak detection method alternatively or additionally includes determining a signature parameter for the liquid of interest, wherein the signature parameter comprises a viscosity difference between the liquid of interest and the other liquid and is determined using the air pressure in the capillary tube; and distinguishing the liquid of interest from the other liquid using a capillary rise differentiation of the liquid of interest from the other liquid, wherein the capillary rise differentiation of the liquid of interest from the other liquid is based on the signature parameter.

[0021] In some example embodiments, the leak detection method alternatively or additionally includes comparing the measured air pressure in the capillary tube with an air pressure threshold; and determining a presence of the liquid of interest when the measured air pressure in the capillary tube is above the air pressure threshold.

[0022] In some example embodiments, the leak detection method alternatively or additionally includes measuring, using a temperature probe of a temperature sensor, a temperature in the spill collector; and varying the air pressure threshold using the measured temperature.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0023] FIG. 1 is a schematic diagram illustrating a liquid cooling system;

[0024] FIG. 2 is a schematic diagram illustrating a leak detection apparatus, in accordance with various embodiments;

[0025] FIG. 3 is a schematic diagram illustrating a leak detection apparatus, in accordance with various embodiments;

[0026] FIG. 4 is a schematic diagram illustrating one or more computing devices, in accordance with various embodiments;

[0027] FIG. 5 illustrates a method in accordance with various embodiments; and,

[0028] FIG. 6 illustrates a method in accordance with various embodiments.

DETAILED DESCRIPTION

[0029] The phrases “in one embodiment,” “according to one embodiment,” “in some embodiments,” “in various embodiments” and the like generally mean that the particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure, and may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

[0030] The word “example” or “exemplary” is used herein to mean “serving as an example,

instance, or illustration” without a limitation. Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. [0031] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that a specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments, or it may be excluded.

[0032] Use of broader terms such as “comprises,” “includes,” and “having” should be understood to provide support for narrower terms such as “consisting of,” “consisting essentially of,” and “comprised substantially of”. Use of the terms “optionally,” “may,” “might,” “possibly,” and the like with respect to any element of an embodiment means that the element is not required, or alternatively, the element is required, both alternatives being within the scope of the embodiment(s). Also, references to examples are merely provided for illustrative purposes, and are not intended to be exclusive.

[0033] The term “computing device” refers to any computer, controller (such as a microcontroller), processor, circuitry, and/or other executor of computer instructions that is embodied in hardware, software, firmware, and/or any combination thereof, that enables access to myriad functionalities associated with one or more mobile device(s), system(s), and/or one or more communications networks. Non-limiting examples of a computing device include a computer, a controller (such as a microcontroller), an application-specific integrated circuit, a field-programmable gate array, a personal computer, a smart phone, a laptop, a fixed terminal, a server, a networking device, a virtual machine, a processor, a plurality of processors electronically coupled to each other and placed in proximity of each other, remote from each other, in a various groups or bundles of one or more processors, and/or forming cloud computing or processing, etc. Other examples of computing devices are provided herein.

[0034] The term “electronically coupled,” “electronically coupling,” “electronically couple,” “in electronic communication with,” or “electronically connected” in the present disclosure refers to two or more elements or components being electronically connected, directly or indirectly. For example, two or more elements or components may be connected through wired means and/or wireless means, such that signals, voltage/current, data, information, or any other electronic signals may be transmitted to and/or received from these elements or components. Electronic connections established via an electrical connector and/or port may refer to wired connections.

[0035] The term “mechanically coupled” in the present disclosure refers to two or more mechanical elements (for example, but not limited to, a frame, a surface, a support unit, a joint, etc.) being physically connected in various ways such as directly, through intermediary elements, and/or using fastener(s), clasps, clamps, joints, pin joint, axle, hinge, adhesive, etc. The term “mechanically coupled” may refer to any of movable, turntable, swiveling, pivoting, fixed, and/or stationary mechanical coupling and/or any other similar type of mechanical coupling. In a non-limiting example, two components are mechanically coupled using a force, such as but not limited to magnetic force, force caused by air pressure, adhesive force, mechanical force, and/or other similar or related forces.

[0036] Battery packs have many technical applications such as in Electric Vehicles (EVs), robotics, household and/or industrial devices, energy distribution units, energy storage or emergency backup units for residential, healthcare, or commercial buildings, data servers, cellular base stations, etc. It is desirable to improve safety and/or reliability of battery packs.

[0037] In some examples, high temperatures may cause safety concerns for battery packs. In some examples, the temperature of a battery pack may increase due to the operation of the battery pack, for example due to current passing through internal conductive paths having electrical resistance in the battery cell, or due to low charging or discharging efficiency of the battery cells in the battery pack. If the increase in the temperature is not regulated, it may lead to a dangerous condition in the

battery pack such as thermal runaway.

[0038] To regulate and keep the temperature within safe operating limits of a battery pack, various types of temperature regulating systems, apparatuses, and methods may be used. For example, a liquid cooling system may be used to regulate the temperature of a battery pack.

[0039] Referring now to FIG. 1 a schematic diagram illustrating a liquid cooling system **100** is provided in accordance with various embodiments of the present disclosure. The liquid cooling system **100** may circulate a coolant, e.g., a liquid coolant, through a tube **104** around the battery cells **106** of a battery pack to regulate the temperature. For example, coolant having a lower temperature is entered in the input **112** of the tube. The coolant may absorb heat from the battery cells **106** and the coolant with a higher temperature may exit from the output **114** of the tube.

[0040] However, there may be a leakage of the coolant from the tube due to a puncture, hole, cut, etc., in the tube. Leakage of the coolant may cause the temperature to increase in the battery pack which may have safety consequences as previously described. In some examples, increased temperatures may reduce efficiency of the battery pack. In some examples, coolant may cause a short circuit between the battery cells, causing increase in the idle load of the battery, which may reduce the efficiency of the battery pack and/or cause fire.

[0041] Various approaches have been used to detect leakage of a coolant. For example, a resistor sensor may be placed in the battery pack where the coolant may be spilled and/or accumulated due to a coolant leakage, for example in a spill collector (e.g., the bottom cover of the battery pack). When the resistor sensor detects a low resistance, it may be an indication of a leakage. However, this approach does not differentiate between a leakage caused by a coolant and condensation in the same location where the resistance sensor is placed.

[0042] Various embodiments of the present disclosure provide methods, systems, and apparatuses to determine a leakage of the coolant while reducing false alarms caused by condensation.

[0043] The coolant may be spilled in the battery pack, for example in a spill collector of the battery pack. In various embodiments, a bottom cover of a battery pack may be the spill collector.

[0044] In various embodiments, a capillary tube is inserted in the battery pack. A capillary tube may be a tube that causes the coolant to rise in the tube due a capillary effect on the coolant in the spill collector. In various embodiments, the capillary effect refers to the rise of a liquid in the capillary tube due to the viscosity of the liquid. In various embodiments, one end of the capillary tube may be placed in the spill collector such that it is immersed in the spilled coolant once a leakage occurs. When a coolant leakage occurs, the coolant may rise in the capillary tube, for example due to the capillary effect.

[0045] In various embodiments, a diameter of the capillary tube is determined such that a capillary effect of interest is seen by liquids having a viscosity similar to the viscosity of the coolant. In example embodiments, because the viscosity of water is different from the viscosity of the coolant, the capillary tube will not have the same capillary effect on water as on the coolant. Hence, in example embodiments, water (for example water caused by condensation in the spill collector) will not have the same rise in the capillary tube as the coolant.

[0046] In various embodiments, due the capillary effect on the coolant, a leakage of coolant is determined and distinguished with a accumulated water in the spill collector. In various embodiments, the capillary effect on the coolant, due to the viscosity of the coolant, is determined to be a signature capillary effect of the coolant. Various embodiments determine a leakage of the coolant using the signature capillary effect of the coolant. For example, when there is water accumulation in the spill collector, there may be a low rise of water, or a negligible rise of water compared to a rise of coolant in the capillary tube.

[0047] In various embodiments, a pressure sensor determines the air pressure created in the capillary tube in case of a rise of a liquid in the capillary tube. In various embodiments, the air pressure is measured and used to determine whether a liquid risen in the capillary tube is a coolant, or another liquid with a different viscosity, such as water. In various embodiments, the pressure

sensor is electronically coupled to one or more computing devices configured to determine a coolant leakage, generate an alarm and/or a coolant leakage detection signal when a rise of a coolant is detected in the capillary tube. By doing so, example embodiments distinguish between a leakage of coolant and an accumulation of other liquids, such as water, in the spill collector. [0048] In various embodiments, a leakage of the coolant may be detected and/or differentiated from an accumulation of other liquids in the spill collectors, such as wax, melted wax, adhesives, etc., that may be used in the manufacturing of the battery packs.

[0049] Various embodiments of the present disclosure may be used to detect a liquid of interest. In various embodiments, a radius of the capillary tube is determined for the liquid of interest and using the viscosity of the liquid of interest, such that an amount of the rise by the liquid of interest in the capillary tube is unique for the liquid of interest. Consequently, in example embodiments, an air pressure created in the capillary tube as a result of the rise of the liquid in the capillary tube is unique for the liquid of interest. In example embodiments, by using the unique air pressure created in the capillary tube consequent to a rise of the liquid of interest in the capillary tube, a presence of the liquid of interest in the spill collector is determined.

[0050] In various embodiments, a radius for the capillary tube is selected such that a capillary rise (h) for the liquid of interest is above a height threshold. The capillary rise (h) for a liquid is determined according to the following equations:

$$[00001] h = (2 * T * \cos(\theta)) / (\rho * g * r)$$

wherein h is the capillary rise, T is a surface tension of the liquid, r is radius of the capillary tube, ρ is a density of the liquid, g is the acceleration due to gravity, and θ is an angle of contact.

[0051] In example embodiments, each liquid may have a unique capillary rise for a given temperature. For example, the viscosity of a liquid may change as the temperature of the liquid varies. In various embodiments, the change of the viscosity due to the variation in temperature is taken into account in detection of the liquid of interest.

[0052] In various embodiments, the detection of the liquid of interest includes thermal compensation. For example, the determination of the liquid of interest is compensated based on the temperature. In various embodiments, when the measured air pressure in the capillary sensor is above a threshold, a presence of the liquid of interest is determined. In various embodiments, the pressure threshold for determining the presence of the liquid of interest is adjusted based on the temperature. For example, the pressure threshold is increased when the temperature of the liquid in the spill collector increases, and the pressure threshold is decreased when the temperature of the liquid in the spill collector decreases.

[0053] Referring now to FIG. 2 a schematic diagram illustrating a leak detection apparatus **200** is provided in accordance with various embodiments of the present disclosure.

[0054] In various embodiments, the leak detection apparatus **200** includes a spill collector **224**. The spill collector **224** may be configured to collect a liquid of interest from a leakage of the liquid of interest. For example, the liquid of interest may be a coolant of a battery pack as previously described, and the spill collector **224** may be the bottom cover of the battery pack.

[0055] In various embodiments, the leak detection apparatus **200** includes a capillary tube **218**. The capillary tube **218** includes a first end **234** configured to immerse in the liquid of interest in the spill collector **224**. For example, when coolant has leaked and accumulated in the spill collector **224**, the first end **234** will be immersed in the coolant.

[0056] In various embodiments, the capillary tube **218** includes a second end **236**. The second end **236** may be configured to mechanically couple with a pressure sensor **226**. In various embodiments, the pressure sensor **226** is configured to measure an air pressure of air **220** in the capillary tube above liquid **222**.

[0057] In various embodiments, the capillary tube **218** is placed such that the first end **234** is immersed in a liquid gathered in spill collector **224**. When a liquid gathers in the spill collector **224**, due to viscosity of the liquid, it rises in the capillary tube **218**. This rise of the liquid increases

the air pressure of air **220** above the liquid **222**. In various embodiments, the pressure sensor **226** is mechanically coupled in an airtight manner at the second end **236** of capillary tube **218**.

[0058] In various embodiments, the leak detection apparatus **200** includes one or more computing devices **232**. The one or more computing devices **232** may be electronically coupled with the pressure sensor **226**. In various embodiments, the one or more computing devices **232** are configured to distinguish the liquid of interest from another liquid using the measured air pressure in the capillary tube. For example, due to condensation, water may be generated and accumulated in the spill collector **224**. However, due to different viscosity of water and coolant, they rise to different levels in the capillary tube **218** causing different air pressures in air **220**. For example, coolant may rise higher than water in capillary tube **218** causing a higher air pressure. In various embodiments, one or more computing devices **232** are configured to determine a presence of coolant in the first end **234** of the capillary tube. Hence a coolant leak may be detected when measured air pressure is above an air pressure threshold.

[0059] In various embodiments, the one or more computing devices **232** are configured to determine a signature parameter for the liquid of interest. In various embodiments, the signature parameter includes and/or is based on a viscosity of the liquid of interest. In various embodiments, the signature parameter includes and/or is based on a difference between the liquid of interest (e.g., coolant) and the other liquid (e.g., water) and is determined using the air pressure in the capillary tube.

[0060] In various embodiments, the one or more computing devices **232** are configured to distinguish the liquid of interest from the other liquid using a capillary rise differentiation of the liquid of interest from the other liquid. In various embodiments, the capillary rise differentiation of the liquid of interest from the other liquid is determined using the measured air pressure and/or a variation in the measure air pressure. In various embodiments, the capillary rise differentiation of the liquid of interest from the other liquid is based on the signature parameter. In various embodiments, the signature parameter of the liquid of interest is determined using the viscosity of the liquid of interest.

[0061] In various embodiments, the one or more computing devices **232** are configured to compare the measured air pressure in the capillary tube with an air pressure threshold. The one or more computing devices **232** may be configured to determine a presence of the liquid of interest when the measured air pressure in the capillary tube **218** is above the air pressure threshold.

[0062] In various embodiments, the leak detection apparatus **200** includes a temperature sensor **228**. The temperature sensor **228** may include a temperature probe **230**. The temperature probe **230** may be configured to engage with, or in contact with, or be inserted in at least one of the liquid of interest or the other liquid in the spill collector **224**. In various embodiments, the temperature sensor **228** is configured to measure a temperature in the spill collector **224** using the temperature probe **230**. For example, the temperature sensor **228** measures the temperature of the coolant and/or water accumulated in the spill collector **224**.

[0063] The viscosity of a liquid may vary based on the temperature of the liquid. For example, the viscosity of a liquid may decrease when the temperature of the liquid increases and vice versa. Variation in viscosity, may change the capillary effect on the liquid. For example, a liquid with lower viscosity may rise higher in the capillary tube than when the temperature is lower.

[0064] To account for the variation of viscosity and capillary effect based on temperature, in various embodiments, the one or more computing devices **232** are further configured to vary the air pressure threshold using the measured temperature. For example, the one or more computing devices **232** are configured to increase the air pressure threshold when the measured temperature is above a temperature threshold. In various embodiments, the one or more computing devices **232** are configured to decrease the air pressure threshold when the measured temperature is below the temperature threshold.

[0065] Referring now to FIG. **3** a schematic diagram illustrating leak detection system **300** is

provided in accordance with various embodiments of the present disclosure.

[0066] In various embodiments, the leak detection system **300** determines a leak in a battery pack, for example a battery pack having liquid cooling system **100** with reference to FIG. **1**. In various embodiments, the leak detection system **300** includes a spill collector. In various embodiments, the spill collector is a bottom cover **304** of the battery pack. For example, the battery pack may be used in an electric vehicle and the bottom cover **304** is a bottom cover of the battery pack of the electric vehicle.

[0067] In various embodiments, the spill collector is configured to collect a liquid of interest from a leakage of the liquid of interest. For example, the liquid of interest is the coolant used to reduce and/or regulate the temperature of a battery pack as previously described. In case of a leakage of the coolant, it is accumulated in the spill collector **306**.

[0068] In various embodiments, the leak detection system **300** includes a capillary tube **218**. The capillary tube may include a first end configured to immerse in the liquid of interest in the spill collector and a second end configured to mechanically couple with a pressure sensor **308**. In various embodiments, the pressure sensor is configured to measure an air pressure in the capillary tube.

[0069] In various embodiments, one or more computing devices are electronically coupled with the pressure sensor. The one or more computing devices may be configured to distinguish the liquid of interest from another liquid using the measured air pressure in the capillary tube. For example, the one or more computing devices may distinguish coolant from water (e.g., generated due to condensation in the spill collector **306**), melted wax and/or melted adhesive, or any other liquid that may be generated in the spill collector **306**. In various embodiments, the one or more computing devices may use any of the techniques or methods described above to determine a presence of the liquid of interest and/or distinguish the liquid of interest from another liquid in the spill collector **306**.

[0070] Referring now to FIG. **4** a schematic diagram illustrating one or more computing device **406** in communication with pressure sensor **402** and/or temperature sensor **416** is provided in accordance with various embodiments of the present disclosure.

[0071] In various embodiments, some or all of the one or more computing device **406** may be placed locally or in proximity with pressure sensor **402** and/or temperature sensor **416** (for example within a same housing) or remotely from pressure sensor **402** and/or temperature sensor **416** (for example outside the housing). In various embodiments, the one or more computing device **406** may be placed inside another housing.

[0072] In various embodiments, the one or more computing device **406** is in communication with pressure sensor **402** and/or temperature sensor **416** or various other components of the leak detection apparatus or system such as a display, a vehicle control system, an alert system, etc., using communication interfaces **412** and over a wired and/or wireless medium **414**. A wired medium **414** may be via any physical coupling of conductive materials, such as using wires, slip rings, linear sliding electrical connections, etc. A wireless medium **414** may include at least one of general packet radio service (GPRS), Universal Mobile Telecommunications System (UMTS), Code Division Multiple Access 2000 (CDMA2000), CDMA2000 1× (1×RTT), Wideband Code Division Multiple Access (WCDMA), Global System for Mobile Communications (GSM), Enhanced Data rates for GSM Evolution (EDGE), Time Division-Synchronous Code Division Multiple Access (TD-SCDMA), Long Term Evolution (LTE), Evolved Universal Terrestrial Radio Access Network (E-UTRAN), Evolution-Data Optimized (EVDO), High Speed Packet Access (HSPA), High-Speed Downlink Packet Access (HSDPA), IEEE 802.11 (Wi-Fi), Wi-Fi Direct, 802.16 (WiMAX), ultra-wideband (UWB), infrared (IR) protocols, near field communication (NFC) protocols, Wibree, Bluetooth protocols, Zigbee, wireless universal serial bus (USB) protocols, and/or any other wireless

[0073] In general, the terms computing apparatus, computer, system, device, entity, and/or similar

words used herein interchangeably can refer to, for example, one or more computers, computing entities, desktops, mobile phones, tablets, notebooks, laptops, distributed systems, kiosks, input terminals, servers or server networks, blades, gateways, switches, processing devices, processing entities, controllers, control systems, set-top boxes, relays, routers, network access points, base stations, the like, and/or any combination of devices or entities adapted to perform the functions, operations, and/or processes described herein. Such functions, operations, and/or processes can include, for example, transmitting, receiving, operating on, processing, displaying, storing, determining, creating/generating, monitoring, evaluating, comparing, and/or similar terms used herein interchangeably. In one embodiment, these functions, operations, and/or processes can be performed on data, content, information, and/or similar terms used herein interchangeably. The one or more computing device **406** can include any computing device including, for example, a mobile device handling and/or processing apparatus configured to perform one or more steps/operations of one or more method or techniques described herein. In some embodiments, the one or more computing device **406** can include and/or be in association with one or more programmable logic controller (PLC), desktop computer(s), laptop(s), server(s), cloud computing platform(s), controller systems, and/or the like. In some example embodiments, the one or more computing device **406** can be configured to receive and/or transmit image processing instructions, data, and/or the like between the one or more leak detection apparatus or systems and/or their components to perform one or more steps/operations of one or more leak detection handling and/or processing techniques or methods described herein.

[0074] The one or more computing device **406** can include, or be in communications with, one or more processing elements **408** (also referred to as processors, processing circuitry, digital circuitry, and/or similar terms used herein interchangeably) that communicate with other elements within the one or more computing device **406** via a bus, for example. As will be understood, the processing elements **408** can be embodied in a number of different ways.

[0075] For example, the processing elements **408** can be embodied as one or more complex programmable logic devices (CPLDs), microprocessors, multi-core processors, coprocessing entities, application-specific instruction-set processors (ASIPs), microcontrollers, and/or controllers. Further, the processing elements **408** can be embodied as one or more other processing devices or circuitry. The term circuitry can refer to an entirely hardware embodiment or a combination of hardware and computer program products. Thus, the processing elements **408** can be embodied as integrated circuits, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), programmable logic arrays (PLAs), hardware accelerators, digital circuitry, and/or the like.

[0076] As will therefore be understood, the processing elements **408** can be configured for a particular use or configured to execute instructions stored in volatile or non-volatile media or otherwise accessible to the processing elements **408**. As such, whether configured by hardware or computer program products, or by a combination thereof, the processing elements **408** can be capable of performing steps or operations according to embodiments of the present disclosure when configured accordingly.

[0077] In one embodiment, the one or more computing device **406** can further include, or be in communication with, one or more memory elements **410**. The one or more memory elements **410** can include non-volatile and/or volatile media. The memory elements **410**, for example, can include non-volatile media (also referred to as non-volatile storage, memory, memory storage, memory circuitry and/or similar terms used herein interchangeably). In one embodiment, the non-volatile storage or memory can include one or more non-volatile storage or memory media, including, but not limited to, hard disks, ROM, PROM, EPROM, EEPROM, flash memory, MMCs, SD memory cards, Memory Sticks, CBRAM, PRAM, FeRAM, NVRAM, MRAM, RRAM, SONOS, FJG RAM, Millipede memory, racetrack memory, and/or the like.

[0078] As will be recognized, the non-volatile storage or memory media can store databases,

database instances, database management systems, data, applications, programs, program modules, scripts, source code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like. The term database, database instance, database management system, and/or similar terms used herein interchangeably can refer to a collection of records or data that is stored in a computer-readable storage medium using one or more database models, such as a hierarchical database model, network model, relational model, entity-relationship model, object model, document model, semantic model, graph model, and/or the like.

[0079] In addition, or alternatively, the memory elements **410** can include volatile memory. For example, the one or more computing device **406** can further include, or be in communication with, volatile media (also referred to as volatile storage memory, memory storage, memory circuitry and/or similar terms used herein interchangeably). In one embodiment, the volatile storage or memory can also include one or more volatile storage or memory media, including, but not limited to, RAM, DRAM, SRAM, FPM DRAM, EDO DRAM, SDRAM, DDR SDRAM, DDR2 SDRAM, DDR3 SDRAM, RDRAM, TTRAM, T-RAM, Z-RAM, RIMM, DIMM, SIMM, VRAM, cache memory, register memory, and/or the like.

[0080] As will be recognized, the volatile storage or memory media can be used to store at least portions of the databases, database instances, database management systems, data, applications, programs, program modules, scripts, source code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like being executed by, for example, the processing elements **408**. Thus, the databases, database instances, database management systems, data, applications, programs, program modules, scripts, source code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like can be used to control certain aspects of the operation of the one or more computing device **406** with the assistance of the processing elements **408** and operating system.

[0081] As indicated, in one embodiment, the one or more computing device **406** can also include one or more communication interfaces **412** for communicating with various computing entities, such as by communicating data, content, information, and/or similar terms used herein interchangeably that can be transmitted, received, operated on, processed, displayed, stored, and/or the like. Such communication can be executed using a wired data transmission protocol, such as fiber distributed data interface (FDDI), digital subscriber line (DSL), Ethernet, asynchronous transfer mode (ATM), frame relay, data over cable service interface specification (DOCSIS), or any other wired transmission protocol. Similarly, the one or more computing device **406** can be configured to communicate via wireless external communication networks using any of a variety of protocols, such as general packet radio service (GPRS), Universal Mobile Telecommunications System (UMTS), Code Division Multiple Access 2000 (CDMA2000), CDMA2000 1× (1×RTT), Wideband Code Division Multiple Access (WCDMA), Global System for Mobile Communications (GSM), Enhanced Data rates for GSM Evolution (EDGE), Time Division-Synchronous Code Division Multiple Access (TD-SCDMA), Long Term Evolution (LTE), Evolved Universal Terrestrial Radio Access Network (E-UTRAN), Evolution-Data Optimized (EVDO), High Speed Packet Access (HSPA), High-Speed Downlink Packet Access (HSDPA), IEEE 802.11 (Wi-Fi), Wi-Fi Direct, 802.16 (WiMAX), ultra-wideband (UWB), infrared (IR) protocols, near field communication (NFC) protocols, Wibree, Bluetooth protocols, Zigbee, wireless universal serial bus (USB) protocols, and/or any other wireless protocol.

[0082] The one or more computing device **406** may perform various steps of various methods for detecting and/or distinguishing a leakage of the liquid of interest, in accordance with various embodiments of the present disclosure.

[0083] Referring now to FIG. 5 a schematic diagram illustrating a method **500** is provided in accordance with various embodiments of the present disclosure.

[0084] In various embodiments, in block **502**, method **500** measures an air pressure in a capillary tube, for example capillary tube **218** or capillary tube **316**. In various embodiments, method **500**

measures the air pressure using a pressure sensor, for example pressure sensor **226** or pressure sensor **308** as previously described.

[0085] In various embodiments, a first end of the capillary tube is configured to immerse in a liquid of interest in a spill collector, for example spill collector **224** or spill collector **306**. In various embodiments, the spill collector is configured to collect a liquid of interest from a leakage of the liquid of interest. For example, the spill collector is configured to collect coolant from a leakage of a coolant in a battery pack as for example shown in FIG. **1**.

[0086] In various embodiments, the pressure sensor is mechanically coupled to a second end of the capillary tube. In various embodiments, the pressure sensor is mechanically coupled in an airtight way to the second end of the capillary tube.

[0087] In various embodiments, in block **504**, method **500** distinguishes the liquid of interest from another liquid using the measured air pressure in the capillary tube using the measured air pressure in the capillary tube. In various embodiments, the method **500** uses the one or more computing devices electronically coupled with the pressure sensor to distinguish the liquid of interest from another liquid.

[0088] Referring now to FIG. **6** a schematic diagram illustrating a method **600** is provided in accordance with various embodiments of the present disclosure.

[0089] In various embodiments, in block **602**, the method **600** determines a signature parameter for the liquid of interest. The signature parameter may include a viscosity difference between the liquid of interest and the other liquid and is determined using the air pressure in the capillary tube.

[0090] In various embodiments, in block **604**, the method **600** distinguishes the liquid of interest from the other liquid using a capillary rise differentiation of the liquid of interest from the other liquid. The capillary rise differentiation of the liquid of interest from the other liquid may be based on the signature parameter.

[0091] In various embodiments, in block **606**, the method **600** compares the measured air pressure in the capillary tube with an air pressure threshold. In various embodiments, at block **608**, the method **600** determines a presence of the liquid of interest when the measured air pressure in the capillary tube is above the air pressure threshold.

[0092] In various embodiments, in block **610**, the method **600** measures, using a temperature probe of a temperature sensor, a temperature in the spill collector at block **610**. In various embodiments, in block **612** the method **600** varies the air pressure threshold using the measured temperature at block **612**.

[0093] In various embodiments, method **600** increases the air pressure threshold when the measured temperature is above a temperature threshold. In various embodiments, method **600** decreases the air pressure threshold when the measured temperature is below the temperature threshold.

[0094] In various embodiments, method **600** uses one or more computing devices to perform various steps, such as determining a signature parameter, distinguishing the liquid of interest from another liquid, comparing the measure air pressure with the air pressure threshold, determining the presence of the liquid of interest, varying the air pressure threshold, generating an output and/or an alarm, etc.

[0095] In various embodiments, method **600** generates an alarm when the presence of the liquid of interest is determined. In various embodiments the liquid of interest is a coolant of a battery pack. In various embodiments, the spill collector is a bottom cover of the battery pack. In various embodiments, the other liquid is at least one of water generated by condensation, melted wax, or melted adhesive in the spill collector.

[0096] Although the example methods depict a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the routine. In other examples, different components of an example device or

system that implements the routine may perform functions at substantially the same time or in a specific sequence.

[0097] Many modifications and other embodiments will come to mind to one skilled in the art to which this disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A leak detection apparatus comprising: a spill collector configured to collect a liquid of interest from a leakage of the liquid of interest; a capillary tube comprising: a first end configured to immerse in the liquid of interest in the spill collector; and a second end configured to mechanically couple with a pressure sensor, the pressure sensor configured to measure an air pressure in the capillary tube; and one or more computing devices electronically coupled with the pressure sensor, the one or more computing devices configured to distinguish the liquid of interest from another liquid using the measured air pressure in the capillary tube.
2. The leak detection apparatus of claim 1, wherein the one or more computing devices are further configured to: determine a signature parameter for the liquid of interest, wherein the signature parameter comprises a viscosity difference between the liquid of interest and the other liquid and is determined using the measured air pressure in the capillary tube; and distinguish the liquid of interest from the other liquid using a capillary rise differentiation of the liquid of interest from the other liquid, wherein the capillary rise differentiation of the liquid of interest from the other liquid is based on the signature parameter.
3. The leak detection apparatus of claim 1, wherein the one or more computing devices are configured to: compare the measured air pressure in the capillary tube with an air pressure threshold; and determine a presence of the liquid of interest when the measured air pressure in the capillary tube is above the air pressure threshold.
4. The leak detection apparatus of claim 3, further comprising a temperature sensor, wherein the temperature sensor comprises a temperature probe configured to engage with at least one of the liquid of interest or the other liquid in the spill collector, the temperature sensor configured to measure a temperature in the spill collector using the temperature probe.
5. The leak detection apparatus of claim 4, wherein the one or more computing devices are further configured to vary the air pressure threshold using the measured temperature.
6. The leak detection apparatus of claim 4, wherein the one or more computing devices are further configured to: increase the air pressure threshold when the measured temperature is above a temperature threshold; and decrease the air pressure threshold when the measured temperature is below the temperature threshold.
7. The leak detection apparatus of claim 3, wherein the one or more computing devices are configured to generate an alarm when the presence of the liquid of interest is determined.
8. The leak detection apparatus of claim 1, wherein the liquid of interest is a coolant of a battery pack, the spill collector is a bottom cover of the battery pack, and wherein the other liquid is at least one of water generated by condensation, melted wax, or melted adhesive in the spill collector.
9. A coolant leak detection system, comprising: a spill collector configured to collect a coolant leaked from a battery pack; a capillary tube comprising: a first end configured to immerse in the coolant when leaked in the spill collector; and a second end configured to mechanically couple with a pressure sensor, the pressure sensor configured to measure an air pressure in the capillary tube above the coolant; and one or more computing devices electronically coupled with the

pressure sensor, the one or more computing devices configured to distinguish the coolant from another liquid in the spill collector using the measured air pressure in the capillary tube.

10. The coolant leak detection system of claim 9, wherein the one or more computing devices are further configured to: determine a signature parameter for the coolant, wherein the signature parameter comprises a viscosity difference between the coolant and the other liquid and is determined using the measured air pressure in the capillary tube; and distinguish the coolant from the other liquid using a capillary rise differentiation of the coolant from the other liquid, wherein the capillary rise differentiation of the coolant from the other liquid is based on the signature parameter.

11. The coolant leak detection system of claim 9, wherein the one or more computing devices are configured to: compare the measured air pressure in the capillary tube with an air pressure threshold; and determine a presence of the coolant when the measured air pressure in the capillary tube is above the air pressure threshold.

12. The coolant leak detection system of claim 11, further comprising a temperature sensor, wherein the temperature sensor comprises a temperature probe configured to engage with at least one of the coolant or the other liquid in the spill collector, the temperature sensor configured to measure a temperature in the spill collector using the temperature probe.

13. The coolant leak detection system of claim 12, wherein the one or more computing devices are further configured to vary the air pressure threshold using the measured temperature.

14. The coolant leak detection system of claim 12, wherein the one or more computing devices are further configured to: increase the air pressure threshold when the measured temperature is above a temperature threshold; and decrease the air pressure threshold when the measured temperature is below the temperature threshold.

15. The coolant leak detection system of claim 11, wherein the one or more computing devices are configured to generate an alarm when the presence of the coolant is determined.

16. The coolant leak detection system of claim 9, wherein the coolant is a coolant of a battery pack and the spill collector is a bottom cover of the battery pack, and wherein the other liquid is at least one of water generated by condensation, melted wax, or melted adhesive in the spill collector.

17. A leak detection method comprising: measuring, by a pressure sensor, an air pressure in a capillary tube, wherein a first end of the capillary tube is configured to immerse in a liquid of interest in a spill collector configured to collect the liquid of interest from a leakage of the liquid of interest, and wherein the pressure sensor is mechanically coupled to a second end of the capillary tube; and distinguishing, by one or more computing devices electronically coupled with the pressure sensor, the liquid of interest from another liquid using the measured air pressure in the capillary tube.

18. The leak detection method of claim 17, further comprising: determining a signature parameter for the liquid of interest, wherein the signature parameter comprises a viscosity difference between the liquid of interest and the other liquid and is determined using the air pressure in the capillary tube; and distinguishing the liquid of interest from the other liquid using a capillary rise differentiation of the liquid of interest from the other liquid, wherein the capillary rise differentiation of the liquid of interest from the other liquid is based on the signature parameter.

19. The leak detection method of claim 17, further comprising: comparing the measured air pressure in the capillary tube with an air pressure threshold; and determining a presence of the liquid of interest when the measured air pressure in the capillary tube is above the air pressure threshold.

20. The leak detection method of claim 19, further comprising: measuring, using a temperature probe of a temperature sensor, a temperature in the spill collector; and varying the air pressure threshold using the measured temperature.
